Stat Comput (2014) 24:725–738
DOI 10.1007/s11222-013-9398-0

Random effects selection in generalized linear mixed models via shrinkage penalty function

Jianxin Pan · Chao Huang

Received: 5 September 2012 / Accepted: 6 April 2013 / Published online: 8 May 2013
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Abstract In this paper, we discuss the selection of random effects within the framework of generalized linear mixed models (GLMMs). Based on a reparametrization of the covariance matrix of random effects in terms of modified Cholesky decomposition, we propose to add a shrinkage penalty term to the penalized quasi-likelihood (PQL) function of the variance components for selecting effective random effects. The shrinkage penalty term is taken as a function of the variance of random effects, initiated by the fact that if the variance is zero then the corresponding variable is no longer random (with probability one). The proposed method takes the advantage of a convenient computation for the PQL estimation and appealing properties for certain shrinkage penalty functions such as LASSO and SCAD. We propose to use a backfitting algorithm to estimate the fixed effects and variance components in GLMMs, which also selects effective random effects simultaneously. Simulation studies show that the proposed approach performs quite well in selecting effective random effects in GLMMs. Real data analysis is made using the proposed approach, too.

Keywords Generalized linear mixed models · Modified Cholesky decomposition · Penalized quasi-likelihood · Penalty function · Random effects

1 Introduction

In statistical modeling, many potential covariates may be included into statistical models at an initial stage, forming a high-dimensional problem when making statistical inferences based on the models. However, some of the covariates may not necessarily make contributions to the models under consideration. Hence, selecting significant covariates from a number of candidates becomes extremely important and necessary. Early literature work on variable selection includes full screening search such as Mallow's C_p, AIC and BIC methods and optimal subset search like backward-, forward- and stepwise-selection methods. It is well known that those methods are either computationally intensive or statistically unreliable (Breiman 1996).

Variable selection via penalty functions was developed in the last decade. In those methods, certain penalty functions of regression coefficients are added to the residual sum of squares or subtracted from the log-likelihood function, then minimization or maximization of the penalized objective functions with respect to regression coefficients leads to penalized least square or penalized likelihood estimates, which helps to remove unnecessary or insignificant covariates from the models. Ridge regression and bridge regression methods with L_2 penalty (Hoerl and Kennard 1970a, 1970b) and L_q $(q \geq 0)$ penalty (Frank and Friedman 1993) were discussed as such examples. The least absolute shrinkage and selection operator (LASSO), which shrinks some coefficients and sets others to 0 with L_1 penalty, was proposed by Tibshirani (1996). This penalty function retains good features of both subset selection and ridge regression. Zou (2006) proposed adaptive LASSO, in which adaptive weights are used in LASSO penalties in order to improve the consistency of LASSO. Another penalty function, called smoothly clipped absolute deviation (SCAD) procedure, was proposed by Fan and Li (2001), providing a continuous differentiable penalty function. The SCAD can be viewed as a modification of LASSO and enjoys the properties of continuity, sparsity and unbiasedness. With such

J. Pan (✉) · C. Huang
School of Mathematics, The University of Manchester, Oxford Road, Manchester M13 9PL, United Kingdom
e-mail: jianxin.pan@manchester.ac.uk

penalty functions, non-informative or insignificant variables are removed from the model by shrinking the associated regression coefficients towards zero, but significant variables are retained in the model with little or no shrinkage. Compared to other variable selection methods, shrinkage penalty methods have the advantage of selecting important covariates and estimating the associated parameters, simultaneously, and save considerable computing times.

Generalized linear mixed models (GLMMs) were proposed for continuous and discrete longitudinal/clustered data analysis, which nowadays have become a standard tool to handle heterogeneous and clustered data. GLMMs incorporate random effects into the linear predictor of generalized linear models, so that heterogeneity across subjects can be taken into account. In the literature, many authors including Stiratelli et al. (1984), Zeger et al. (1988), Schall (1991), Breslow and Clayton (1993) and Pan and Thompson (2007) among others, have contributed to the modeling development. The most commonly used estimation method is the penalized quasi-likelihood (PQL) estimation proposed by Breslow and Clayton (1993). The key idea of the PQL method is to utilize Laplace's approximation to approximate the integrated likelihood. The main advantage of the PQL method lies in that it can be easily fulfilled by iteratively fitting a linear mixed model to a modified working response (Breslow and Clayton 1993).

Variable selection in GLMMs has been studied by many authors in recent years. Among those, Schelldorfer and Bühlmann (2011) and Groll and Tutz (2013) considered the selection of fixed effects in GLMMs using L_1-penalization. Groll and Tutz (2012) applied boosting techniques to the selection of fixed effects in GLMMs. Ibrahim et al. (2010) proposed a variable selection method within the framework of a general class of mixed-effects models by the use of a penalized likelihood function with SCAD and adaptive LASSO penalties, where a model selection criterion called IC_Q criterion is used to select the tuning parameter. In contrast, the selection of random effects in GLMMs has received relatively little attention. Note that the selection of effective random effects is very important, because redundant random effects may lead to a singular covariance matrix of random effects and cause unstable computation problems of parameter estimation. Chen and Dunson (2003) and Kinney and Dunson (2007) proposed random effects selection methods based on hierarchical Bayesian models where random effects are selected using a Markov chain Monte Carlo (MCMC) algorithm. Fahrmeir et al. (2010) gave a detailed review of Bayesian approaches for the selection of random effects. When GLMMs reduce to linear mixed models (LMMs), Bondell et al. (2010) proposed a variable selection method for both fixed and random effects using the EM algorithm. For LMMs, very recently Ahn et al. (2012) and Lin et al. (2012) proposed approaches on random effects selection using moment-based method and two-stage method, respectively. Fan and Li (2012) proposed a group variable selection strategy to simultaneously select and estimate random effects, by using a proxy matrix to replace the unknown covariance matrix of random effects. Note that their method is restricted to LMMs and the penalties are taken as a function of random effects.

In this paper, we focus on the selection of random effects within the framework of GLMMs. Based on a reparametrization of the covariance matrix of random effects in terms of a modified Cholesky decomposition, we propose to add a shrinkage penalty term to the PQL function of the variance components for selecting effective random effects. The shrinkage penalty term is taken as a function of the variances of random effects, initiated by the fact that if the variance is equal to zero then the corresponding variable is no longer random in the sense of probability being equal to 1 and can thus be treated as a constant. The proposed method takes the advantage of a convenient computation for the PQL estimation and appealing properties for certain shrinkage penalty functions such as LASSO and SCAD. We then propose to use a backfitting algorithm to estimate the fixed effects and variance components in GLMMs, which also selects effective random effects simultaneously. The organization of this paper is outlined as follows. In Sect. 2, the methodology and the algorithm for selecting effective random effects are discussed. In Sect. 3, simulation studies are carried out to access the performance of the proposed method. A real data analysis is provided in Sect. 4. Further discussions and concluding remarks are given in Sect. 5.

2 Methodology and algorithm

Let y_{ij} be the jth of n_i response measurements on the ith of n subjects. Assume $\mu_{ij} = \mathrm{E}(y_{ij}|b_i)$ is the conditional expectation of the measurement y_{ij}, given the q-dimensional random effects b_i. GLMMs can be defined by

$$\begin{aligned} &g(\mu_{ij}) = \eta_{ij} = x_{ij}^T\beta + z_{ij}^T b_i, \\ &\mathrm{var}(y_{ij}|b_i) = a(\phi)v(\mu_{ij}), \end{aligned} \tag{2.1}$$

where β is the p-dimentional vector of fixed effects, x_{ij} and z_{ij} are covariates corresponding to β and b_i, respectively, $g(\cdot)$ is a known monotone and differenciable link function, $v(\cdot)$ and $a(\cdot)$ are known specific functions in the variance, and ϕ is a nuisance dispersion parameter.

The random effects b_i are assumed to follow a normal distribution $b_i \sim N(0, D)$, where D is an $(q \times q)$ covariance matrix, $D = (d_{ij})_{1\le i,j\le q}$. Based on the modified Cholesky decomposition (Chen and Dunson 2003), the covariance matrix D can be uniquely decomposed into $D = \Lambda\Gamma\Gamma^T\Lambda$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_q)$ is a diagonal matrix with positive elements λ_l proportional to the standard deviation of the random effects b_i, and $\Gamma = (\gamma_{lr})_{1\le l,r\le q}$

is a lower triangular matrix with 1's as its diagonal elements, that is, $\gamma_{ll} = 1$ and $\gamma_{lr} = 0$ for $l = 1, \ldots, q$; $r = l+1, \ldots, q$. The matrix Γ actually determines the correlation of the components of the random effects b_i. Denote $\theta_1 = (\lambda_1, \ldots, \lambda_q)^T$, $\theta_2 = (\gamma_{21}; \gamma_{31}, \gamma_{32}; \ldots; \gamma_{q1}, \ldots, \gamma_{q(q-1)})^T$ and $\theta = (\theta_1^T, \theta_2^T, \phi)^T$.

Conditionally on the random effects b_i, we denote the density function of the response y_{ij} by $f(y_{ij}|b_i; \beta, \phi)$. Let $y_i = (y_{i1}, y_{i2}, \ldots, y_{in_i})^T$ for $i = 1, \ldots, n$. The marginal likelihood function, $L(\beta, \theta)$, that integrates out the unobserved random effects from the joint likelihood function of y_i and b_i, can be expressed as

$$L(\beta,\theta) = \prod_{i=1}^{n} f(y_i;\beta,\theta) = \prod_{i=1}^{n} \int_{b_i} \left\{ \prod_{j=1}^{n_i} f(y_{ij}|b_i;\beta,\phi) \right\} f(b_i;\theta_1,\theta_2)db_i,$$

where $f(b_i;\theta_1,\theta_2) = (2\pi)^{-q/2}|D|^{-1/2}\exp\{-\frac{1}{2}b_i^T D^{-1}b_i\}$. Accordingly, the log-likelihood, $\ell(\beta,\theta)$, has the following form

$$\ell(\beta,\theta) = \sum_{i=1}^{n} \log f(y_i;\beta,\theta) = \sum_{i=1}^{n} \log \int_{b_i} \left\{ \prod_{j=1}^{n_i} f(y_{ij}|b_i;\beta,\phi) \right\} f(b_i;\theta_1,\theta_2)db_i.$$

The integral above may have a closed form when the conditional distribution of y_{ij} given b_i is normal. However, in general this likelihood function is analytically intractable due to the complex form of the functions $\prod_{j=1}^{n_i} f(y_{ij}|b_i; \beta, \phi)$. As an approximation to $\ell(\beta,\theta)$, the PQL approximates the log-likelihood through

$$\ell(\beta,\theta) \approx \sum_{i=1}^{n}\sum_{j=1}^{n_i} \log f(y_{ij}\,|\,b_i;\beta,\phi) - \frac{1}{2}\sum_{i=1}^{n} b_i^T D^{-1} b_i \tag{2.2}$$

Then the PQL estimation can be made by iterating the two steps below. First, for given variance components θ, maximization of $\ell(\beta,\theta)$ with respect to β and b_i is taken, so that the estimates $(\hat\beta, \hat b_i) = (\hat\beta(\theta), \hat b_i(\theta))$ can be obtained in explicit expression forms. In fact, let $\eta_i = (\eta_{i1}, \ldots, \eta_{in_i})^T$, $\mu_i = (\mu_{i1}, \ldots, \mu_{in_i})^T$ and $g'(\mu_i) = (g'(\mu_{i1}), \ldots, g'(\mu_{in_i}))^T$. Then the estimating equations for β and b_i are given by

$$\sum_{i=1}^{n} X_i^T W_i(Y_i - \eta_i) = 0 \quad \text{and} \quad Z_i^T W_i(Y_i - \eta_i) = D^{-1}b_i$$

where $X_i = (x_{i1}, \ldots, x_{in_i})^T$, $Z_i = (z_{i1}, \ldots, z_{in_i})^T$, $Y_i = \eta_i + (y_i - \mu_i)g'(\mu_i)$ is the working responses, and $W_i = \mathrm{diag}(w_{i1}, \ldots, w_{in_i})$ with $w_{ij} = \{a(\phi)v(\mu_{ij})g'(\mu_{ij})^2\}^{-1}$ is the weight matrix. The solutions to the estimating equations are then provided by

$$\hat\beta = \left(\sum_{i=1}^{n} X_i^T V_i^{-1} X_i\right)^{-1} \left(\sum_{i=1}^{n} X_i^T V_i^{-1} Y_i\right), \tag{2.3}$$

$$\hat b_i = DZ_i^T V_i^{-1}(Y_i - X_i\hat\beta), \tag{2.4}$$

where $V_i = W_i^{-1} + Z_i D Z_i^T$. Therefore, as long as a starting value of (β, θ) is given, the above formulaes can be iteratively used to update the estimate of the fixed effects β and the prediction of the random effects b_i.

Second, once the estimate of β and the prediction of b_i are obtained, the log-likelihood function can be further approximated by

$$\ell_2(\hat\beta,\theta) = -\frac{1}{2}\sum_{i=1}^{n}\log|V_i| - \frac{1}{2}\sum_{i=1}^{n}(Y_i - X_i\hat\beta)^T V_i^{-1}(Y_i - X_i\hat\beta)$$

(Breslow and Clayton 1993). An alternative approximation, which corresponds to the restricted maximum likelihood (REML), is given by

$$\ell_{2R}(\hat\beta,\theta) = -\frac{1}{2}\sum_{i=1}^{n}\log|V_i| - \frac{1}{2}\sum_{i=1}^{n}\log\left|X_i^T V_i^{-1} X_i\right| - \frac{1}{2}\sum_{i=1}^{n}(Y_i - X_i\hat\beta)^T V_i^{-1}(Y_i - X_i\hat\beta),$$

aiming to reduce the bias of the estimates of the variance components θ (Patterson and Thompson 1971). Maximization of $\ell_2(\hat\beta,\theta)$ or $\ell_{2R}(\hat\beta,\theta)$ with respect to θ leads to the ML and REML estimate of θ, respectively. Then the estimates of β and θ are iterated until convergence, yielding the PQL estimates of β and θ. See Breslow and Clayton (1993) for more details.

Consider the lth diagonal entry of Λ in $D = \Lambda\Gamma\Gamma^T\Lambda$, say λ_l. It is obvious that if $\lambda_l = 0$ then $d_{ll} = 0$, so that all the entries of the lth row and the lth column in D are equal to zero. In other words, the lth component of the random effects b_i becomes ineffective as long as $d_{ll} = 0$, so that it can be removed from b_i in this case. In the spirit of penalty-based variable selection methods, we propose to impose penalties on λ_l $(l = 1, \ldots, q)$ to either $\ell_2(\hat\beta,\theta)$ or $\ell_{2R}(\hat\beta,\theta)$ when estimating the variance components θ, so as to shrink the variances of ineffective random effects towards zero. In other words, we maximize the following penalized log-likelihood function

$$p\ell_2(\hat\beta,\theta) = \ell_2(\hat\beta,\theta) - np_\psi\big(|\theta_1|\big), \tag{2.5}$$

or its REML version

$$p\ell_{2R}(\hat\beta,\theta) = \ell_{2R}(\hat\beta,\theta) - np_\psi\big(|\theta_1|\big) \tag{2.6}$$

with respect to θ, so that ineffective random effects can be automatically removed from b_i. In (2.5) and (2.6),

$p_\psi(|\theta_1|) = \sum_{l=1}^{q} p_\psi(|\lambda_l|)$ are the penalty functions for $\lambda_1, \ldots, \lambda_q$ with ψ as the tuning parameter, which can be chosen as either LASSO or SCAD penalty functions. For the LASSO penalty, it is given by $p_\psi(|\lambda_l|) = \psi|\lambda_l|$. For the SCAD penalty, it takes the form

$$p_\psi\big(|\lambda_l|\big) = \psi\left\{I\big(|\lambda_l| \le \psi\big) + \frac{(a\psi - |\lambda_l|)_+}{(a-1)\psi} I\big(|\lambda_l| > \psi\big)\right\},$$

for some $a > 2$ and $|\lambda_l| > 0$, where $I(\cdot)$ is the indicator function and $(x)_+ = x$ if $x > 0$ and $(x)_+ = 0$ otherwise. Note that in (2.5) and (2.6) a common tuning parameter ψ is applied to all $\lambda_1, \ldots, \lambda_q$, implying that the random effects are all in a comparable scale. Otherwise, the random effects may need to be standardized first. We also point out that the penalty function in (2.5) or (2.6) is multiplied by the factor n, because in total there are n independent subjects.

In what follows, we discuss how to estimate the variance components $\theta = (\theta_1^T, \theta_2^T, \phi)^T$ based on the penalized log-likelihood function $p\ell_2(\hat\beta, \theta)$ or $p\ell_{2R}(\hat\beta, \theta)$. We first assume that ϕ is known. Then maximizing $p\ell_2(\hat\beta, \theta)$ or $p\ell_{2R}(\hat\beta, \theta)$ with respect to θ_1 and θ_2 leads to the PQL estimates $\hat\theta_1$ and $\hat\theta_2$ of θ_1 and θ_2, respectively. To obtain the estimate of θ_1, we fix θ_2 first and in this case $p\ell_2(\hat\beta, \theta)$ in (2.5) or $p\ell_{2R}(\hat\beta, \theta)$ in (2.6) is simply a function of θ_1 only. We then use Newton-Raphson algorithm to calculate the estimate $\hat\theta_1$. Note that the penalty function $p_\psi(\cdot)$ may not always have continuous second-order derivatives. We therefore propose to replace the penalty function with its quadratic approximation functions (Fan and Li 2001). The details are given below. Let $\theta_1^{(k)}$ and $\theta_1^{(k-1)}$ be the kth and $(k-1)$th iteration estimates of θ_1. Given β and θ_2, the estimate of θ_1 can be iteratively updated using

$$\theta_1^{(k)} = \theta_1^{(k-1)} - \big(\nabla^2\ell_2\big(\theta_1^{(k-1)}\big) + n\Sigma_\psi\big(\theta_1^{(k-1)}\big)\big)^{-1} \cdot \big(\nabla\ell_2\big(\theta_1^{(k-1)}\big) + nU_\psi\big(\theta_1^{(k-1)}\big)\big), \tag{2.7}$$

where $\nabla\ell_2(\theta_1) = \frac{\partial\ell_2}{\partial\theta_1}$ and $\nabla^2\ell_2(\theta_1) = \frac{\partial^2\ell_2}{\partial\theta_1\theta_1^T}$ are the first and second order derivatives of $\ell_2(\beta, \theta)$ with respect to θ_1, and

$$U_\psi\big(\theta_1^{(k-1)}\big) = \Sigma_\psi\big(\theta_1^{(k-1)}\big)\theta_1^{(k-1)}$$

and

$$\Sigma_\psi\big(\theta_1^{(k-1)}\big) = \mathrm{diag}\left(\frac{p'_\psi(|\lambda_1^{(k-1)}|)}{|\lambda_1^{(k-1)}|}, \ldots, \frac{p'_\psi(|\lambda_q^{(k-1)}|)}{|\lambda_q^{(k-1)}|}\right)$$

are the approximated first and second order derivatives of $p_\psi(|\theta_1^{(k-1)}|)$ with respect to θ_1. Alternatively, the REML estimate of θ_1 can be obtained simply by replacing ℓ_2 with ℓ_{2R}.

We now turn to calculate the estimate of θ_2. Suppose that $\hat\theta_1$ is already obtained using the iteration in (2.7). Assume certain components of $\hat\theta_1$ are equal to zero, for example, $\hat\lambda_l = 0$. Then the lth component of the random effects b_i is not effective, so that it should be removed from b_i. As a result, $\gamma_{l1}, \ldots, \gamma_{l(l-1)}$ are removed from θ_2. The remaining components in θ_2 can then be updated using

$$\theta_2^{(k)} = \theta_2^{(k-1)} - \big(\nabla^2\ell_2\big(\theta_2^{(k-1)}\big)\big)^{-1}\big(\nabla\ell_2\big(\theta_2^{(k-1)}\big)\big), \tag{2.8}$$

where $\nabla\ell_2(\theta_2) = \frac{\partial\ell_2}{\partial\theta_2}$ and $\nabla^2\ell_2(\theta_2) = \frac{\partial^2\ell_2}{\partial\theta_2\theta_2^T}$. The derivation of the first- and second-order derivatives in (2.7)–(2.8) is provided in Appendix, where the details of $\Sigma_\psi(\theta_1^{(k-1)})$ and $U_\psi(\theta_1^{(k-1)})$ are presented in Part 1 there. The technical details of the derivatives $\nabla\ell_2(\theta_s^{(k-1)})$ and $\nabla^2\ell_2(\theta_s^{(k-1)})$ $(s = 1, 2)$ are given in Part 2 of Appendix.

At convergence, the covariance matrices of $\hat\theta_1$ and $\hat\theta_2$ can be calculated using the following sandwich formulae (Fan and Li 2001),

$$\mathrm{Cov}(\hat\theta_1) = \big(\nabla^2\ell_2(\hat\theta_1) + n\Sigma_\psi(\hat\theta_1)\big)^{-1}\widehat{\mathrm{Cov}}\big(\nabla\ell_2(\hat\theta_1)\big) \cdot \big(\nabla^2\ell_2(\hat\theta_1) + n\Sigma_\psi(\hat\theta_1)\big)^{-1}$$

and

$$\mathrm{Cov}(\hat\theta_2) = \big(\nabla^2\ell_2(\hat\theta_2)\big)^{-1}\widehat{\mathrm{Cov}}\big(\nabla\ell_2(\hat\theta_2)\big)\big(\nabla^2\ell_2(\hat\theta_2)\big)^{-1}.$$

If ϕ is unknown, it can be estimated iteratively using the residuals. For example, if y_{ij} follow a normal distribution and it is assumed that the predictions of random effects b_i are already obtained, then ϕ can be updated using $\phi^{(k)} = (1/N)\sum_{i=1}^{n}(e_i^{(k)})^T e_i^{(k)}$ where $N = \sum_{i=1}^{n} n_i$ and $e_i^{(k)} = y_i - X_i\hat\beta^{(k)} - Z_i\hat b_i^{(k)}$ are the conditional residuals in the kth iteration. If y_{ij} follow a binomial or a Poisson distribution conditional on the random effects b_i, we may simply take $\phi = 1$ as long as no over- or under-dispersion occurs. In general, the dispersion parameter ϕ can be estimated by the help of the Pearson residuals and the trace of the corresponding hat matrix as the effective degrees of freedom. See Groll and Tutz (2013) for the details.

In summary, we propose to use the algorithm below to calculate the estimates of the fixed effects β and the variance components θ, which selects effective random effects simultaneously.

Algorithm

Step 1. Given initial values of the fixed effects $\beta^{(0)}$ and the variance components $\theta^{(0)} = (\theta_1^{(0)}, \theta_2^{(0)}, \phi^{(0)})$, compute the estimate of fixed effects $\beta^{(1)}$ and the prediction of random effects $b_i^{(1)}$ using (2.3) and (2.4), respectively.

Step 2. Given $\beta^{(1)}$, $\theta_2^{(0)}$ and $\phi^{(0)}$, use (2.7) to update the variance component estimate $\theta_1^{(1)}$. Then, insert $\beta^{(1)}$, $\theta_1^{(1)}$ and $\phi^{(0)}$ to (2.8), and compute the variance component estimate $\theta_2^{(1)}$. Update $\phi^{(0)}$ by $\phi^{(1)}$ if necessary.

Step 3. Repeat Steps 1 and 2 above until convergence.

Note that when y_i are normally distributed, ϕ needs to be estimated. In this case, we first calculate the residuals

$e_i^{(1)} = y_i - X_i\hat{\beta}^{(1)} - Z_i\hat{b}_i^{(1)}$ in Step 1 and then update the estimate $\phi^{(1)}$ using $e_i^{(1)}$. In addition, unless there is some prior information about the covariance matrix D of the random effects b_i, in general, the identity structure can be chosen as the initial value of D. Equivalently, $\theta_1^{(0)} = 0$ and $\theta_2^{(0)} = 0$ can be chosen as the initial values of θ_1 and θ_2.

Note that the tuning parameter ψ in the penalty function $p_\psi(|\theta_1|)$ is generally unknown and needs to be estimated together with other unknown parameters. We propose to use the leave-one-subject-out cross validation (SCV) method to obtain the optimal tuning parameter ψ. In other words, we minimize the SCV criterion

$$\mathrm{SCV}(\psi) = \sum_{i=1}^{n}\big(Y_i^{(-i)} - X_i\hat{\beta}^{(-i)}\big)^T\big[V_i^{(-i)}\big]^{-1} \times \big(Y_i^{(-i)} - X_i\hat{\beta}^{(-i)}\big),$$

with respect to ψ in order to obtain the optimal tuning parameter, where $Y_i^{(-i)} = \eta_i^{(-i)} + (y_i - \mu_i^{(-i)})g'(\mu_i^{(-i)})$, $V_i^{(-i)} = (W_i^{(-i)})^{-1} + Z_i D^{(-i)} Z_i^T$, $D^{(-i)}$ are computed using data without the ith subject, and $\hat{\beta}^{(-i)}$, $\eta_i^{(-i)}$, $\mu_i^{(-i)}$, $W_i^{(-i)}$ are the corresponding vector or matrix estimates after deleting the ith subject.

3 Simulation studies

In this section, simulation studies are carried out to assess the performance of the proposed method for selecting random effects in GLMMs. The studies are made for four cases, i.e., normal, Bernoulli, binomial and Poisson distributions. In each case, we consider two different model scenarios for longitudinal data, of which the first scenario assumes that covariates are time-independent and the second one considers the case for time-dependent covariates. We use LASSO and SCAD penalty functions when selecting effective random effects. Note that for the SCAD penalty we choose the parameter $a = 3.7$, as suggested by Fan and Li (2001).

3.1 Normal distribution

3.1.1 Time-independent covariates

For the normal distribution with time-independent covariates, we generate 100 data sets, of which each contains 100 subjects ($n = 100$) and each subject has 10 repeated measurements ($n_i \equiv 10$). The model considered is given by $\mu_{ij} = \eta_{ij} = x_{ij}^T\beta + z_{ij}^T b_i$, where $x_{ij} \sim N_p(0, \sigma^2 V)$ with $V = (\rho^{|j-k|})_{1\le j,k\le p}$ and $p = 10$, and the true values of σ^2 and ρ are $\sigma^2 = 1$ and $\rho = 0.2, 0.5, 0.8$, respectively. We also assume $z_{ij} = x_{ij}$ and $\mathrm{var}(y_{ij}|b_i) \equiv 1$, meaning $a(\phi) = 1$. The fixed effects β are set to be a 10-dimensional vector with the true value $\beta = (10, 9, 8, 7, 6, 5, 4, 3, 2, 1)$, and the random effects $b_i = (b_{i1}, b_{i2}, \ldots, b_{i,10}) \sim N_{10}(0, D)$ with $D = \mathrm{diag}(3, 2, 1, 0, \ldots, 0)$, in other words, only the first three random effects are effective. The LASSO and SCAD penalties are implemented in our proposed method for selecting random effects and the SCV criterion is used to choose the optimal tuning parameter ψ.

Table 1 summarizes the estimation results for the mean parameter β. In Table 1, both the averaged mean (Mean) and stand deviation (SD) of $\hat{\beta}$ over 100 simulations are reported. It is clear that all the estimates of β are very close to the true parameter values. Also, it is interesting to see that the correlation parameter ρ has little effect on the estimation of β. And there is no much difference in the estimates of β between LASSO and SCAD methods, when the focus is to select effective random effects. This is obvious from (2.5) or (2.6) because the penalty function is only imposed onto the variance components θ_1. For this reason, in all other simulation studies below we omit the report of the estimation of the fixed effects and focus on the selection of random effects.

To measure the effectiveness of random effects selection, motivated by the "Median Ratio of Model Error" (MRME, Fan and Li 2001), we propose to consider the following criterion, called "Average Error" (AE),

$$\mathrm{AE} = \sum_{l=1}^{q}\frac{1}{q}(\lambda_l - \hat{\lambda}_l)^2 + \sum_{l=2}^{q}\sum_{r=1}^{l-1}\frac{1}{q(q-1)/2}(\gamma_{lr} - \hat{\gamma}_{lr})^2,$$

where λ_l and γ_{lr} are the covariance components of the random effects and $\hat{\lambda}_l$ and $\hat{\gamma}_{lr}$ are the associated estimates. The ratio of the AE of the model with either the LASSO or SCAD penalty function to the model without any penalty function is calculated for each simulation, and the median of the ratios over the 100 simulations, called "Median Ratio of Average Error" (MRAE), is used to measure the performance of the proposed random effects selection method.

Table 2 gives the estimation results in terms of MRAE, correct and incorrect number of zeros (C and IC), the proportion of under-fit, correct-fit and over-fit. Here the correct number of zeros means the number of zero-components in average over the 100 simulations that are correctly identified as zero, whilst the incorrect number of zeros means the number of nonzero components in average over the 100 simulations that are incorrectly judged to be zero. For this simulation study, we would expect that the correct number of zeros should be close to 7 as there are 7 zero components, and the incorrect number of zeros should be close to 0. Under-fit in Table 2 represents the case in which some non-zero components are mistakenly judged as zero, whilst over-fit means that some zero components are wrongly identified as nonzero. From Table 2, it is clear that the MRAE value is very small, whatever the correlation coefficient ρ of the covariates is, implying that the proposed method is

Table 1 Mean parameters estimation

	ρ		Fixed effect									
			β_1	β_2	β_3	β_4	β_5	β_6	β_7	β_8	β_9	β_{10}
			10	9	8	7	6	5	4	3	2	1
SCAD	0.2	(Mean)	10.01	9.02	7.99	6.99	6.00	5.00	4.00	3.00	2.00	1.00
		(SD)	0.21	0.16	0.11	0.05	0.04	0.05	0.04	0.04	0.05	0.04
	0.5	(Mean)	10.01	8.97	7.99	7.00	6.00	5.00	3.99	3.01	1.99	1.00
		(SD)	0.17	0.16	0.10	0.05	0.05	0.05	0.05	0.06	0.05	0.04
	0.8	(Mean)	9.99	8.98	8.00	7.02	5.97	5.01	3.99	3.01	2.00	1.00
		(SD)	0.17	0.17	0.14	0.08	0.08	0.09	0.09	0.09	0.09	0.08
LASSO	0.2	(Mean)	10.01	9.03	7.99	6.99	6.00	5.00	4.00	3.00	2.00	1.00
		(SD)	0.21	0.16	0.11	0.05	0.04	0.05	0.04	0.04	0.05	0.04
	0.5	(Mean)	10.01	8.97	7.99	7.00	6.00	5.00	3.99	3.01	1.99	1.00
		(SD)	0.17	0.16	0.11	0.05	0.05	0.05	0.05	0.06	0.05	0.04
	0.8	(Mean)	9.99	8.98	8.00	7.02	5.97	5.01	3.99	3.01	2.00	1.00
		(SD)	0.17	0.17	0.14	0.08	0.08	0.09	0.09	0.09	0.09	0.08

Table 2 Random effects selection

Method	ρ	MRAE	No. of zeros		Proportion of		
			C	IC	Under-fit	Correct-fit	Over-fit
SCAD	0.2	1.31e-2	7.00	0.02	0.02	0.98	0.00
	0.5	6.38e-3	7.00	0.00	0.00	1.00	0.00
	0.8	1.38e-4	7.00	0.05	0.05	0.95	0.00
LASSO	0.2	1.26e-2	7.00	0.01	0.01	0.99	0.00
	0.5	6.07e-3	7.00	0.00	0.00	1.00	0.00
	0.8	1.37e-4	7.00	0.05	0.05	0.95	0.00

able to reduce the estimation error of the variance components of random effects. Also, the proposed random effect selection method behaves very well in terms of selecting effective random effects and dropping false ones. Moreover, despite of occasional under-fit cases, the proposed method has a very high correct-fit probability for both LASSO and SCAD methods. It seems that the incorrect number of zeros and the percentage of under-fit increase with ρ, which reflects the degree of collinearity among covariates.

In Fig. 1, box plots are provided to demonstrate how the proposed random effects selection method works for the case of $\rho = 0.2$ with SCAD penalty. For example, the first three box plots in Fig. 1 show the performance of variance estimates for random effects b_{i1}, b_{i2} and b_{i3} among 100 simulations. As their true variance values are $d_{11} = 3$, $d_{22} = 2$ and $d_{33} = 1$, they are obviously effective random effects. In contrast, random effects $b_{i4}, \ldots, b_{i,10}$ are ineffective because their variances are zero. Figure 1 shows this point very well.

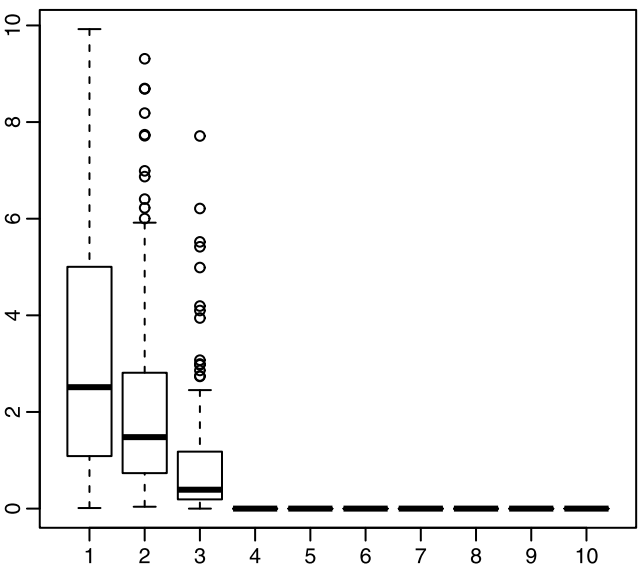


Fig. 1 Box plots of variances estimates of $b_{i1}, b_{i2}, \ldots, b_{i,10}$

Table 3 Random effects selection

Method	n	MRAE	No. of zeros		Proportion of		
			C	IC	Under-fit	Correct-fit	Over-fit
SCAD	100	5.6e-5	4.00	0.03	0.03	0.97	0.00
	200	5.8e-5	4.00	0.02	0.02	0.98	0.00
LASSO	100	5.7e-5	4.00	0.03	0.03	0.97	0.00
	200	6.1e-5	4.00	0.02	0.02	0.98	0.00

We also estimate the dispersion parameter ϕ by using residuals. Note that the true value is $\phi = 1$. The estimates are obtained as 1.04, 1.05 and 1.08 for the cases of $\rho = 0.2, 0.5, 0.8$ using the SCAD penalty, and 1.04, 1.06 and 1.06 for the same values of ρ using the LASSO penalty. These results show that the residuals method for estimating ϕ is fairly acceptable.

3.1.2 Time-dependent covariates

In time-dependent covariates normal case, we use the same model as for the time-independent covariates normal case except that the covariates for fixed effects and random effects are assumed to be $(p-1)$th and $(q-1)$th order polynomials in time, respectively. The measurement times t_{ij} are generated by $t_{ij} = j - 0.5 + e_{ij}$ where $e_{ij} \sim N(0, 0.1)$ are random perturbations. After standardization of t_{ij}, $\tilde{t}_{ij} = (t_{ij} - \bar{t}_{i\cdot})/s(t_{i\cdot})$, where $\bar{t}_{i\cdot}$ and $s(t_{i\cdot})$ represent the sample mean and standard deviation of time points $t_{i1}, t_{i2}, \ldots, t_{in_i}$, the time-dependent covariates involving the $(p-1)$th and $(q-1)$th order polynomials of $\tilde{t}_{ij}$ are set to be covariates: $x_{ij} = (1, \tilde{t}_{ij}, \ldots, \tilde{t}_{ij}^{p-1})$, $z_{ij} = (1, \tilde{t}_{ij}, \ldots, \tilde{t}_{ij}^{q-1})$. For simplicity, we assume $z_{ij} = x_{ij}$ with $p = q = 6$, meaning the 5th order polynomials in time are used to model the data y_{ij}. We also assume $\mathrm{var}(y_{ij}|b_i) \equiv 1$, meaning $a(\phi) = 1$. The random effects are $b_i = (b_{i1}, b_{i2}, \ldots, b_{i6}) \sim N(0, D)$ with $D = \mathrm{diag}(2, 1, 0, 0, 0, 0)$, implying only the intercept and the linear trend of measurement times involve effective random effects in the true model. The true value of the fixed effects β is set to be $\beta = (6, -5, 4, -3, 2, -1)$. For each combination of $n = 100, 200$ and $n_i \equiv 10$, we carried out 100 simulations. For each simulation, the LASSO and SCAD penalty methods are used to select effective random effects.

Similar to Table 2, the MRAE, number of zeros, proportion of under-, over- and correct-fit are reported in Table 3 to assess the performance of the proposed method for random effects selection. From Table 3, it is clear that there is a small number of under-fit cases, leading to small incorrect number of selecting effective random effects. In fact, there are 97 % and 98 % correct selection of effective random effects among the 100-time simulations, and only 3 % and 2 % of under-fit for random effects. Figure 2 provides the box plot for the performance of the proposed random effects selection method for the case with $n = 100$ using SCAD penalty, which again demonstrates the good performance of the proposed method.

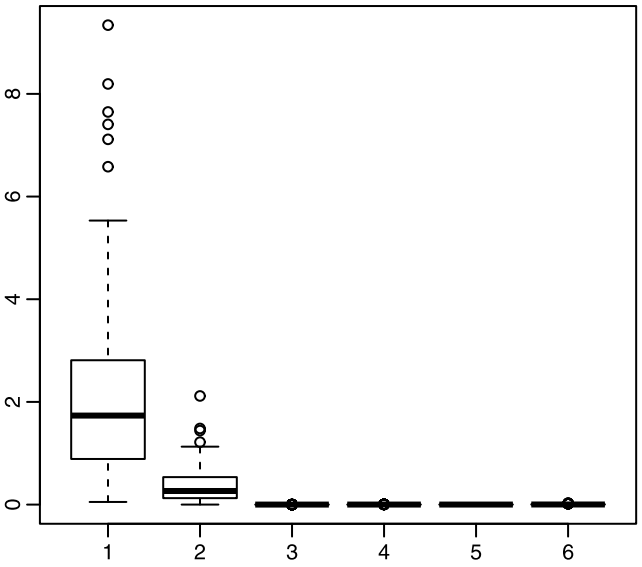


Fig. 2 Box plots of variances estimates of $b_{i1}, b_{i2}, \ldots, b_{i,6}$

Finally, we also give the estimation value of the nuisance parameter ϕ. When $n = 100$, then $\hat{\phi} = 1.05$ for both SCAD and LASSO methods. When n increase to 200, $\hat{\phi} = 1.05$ for the SCAD method and $\hat{\phi} = 1.04$ for the LASSO method. These estimates are quite close to $\phi = 1$ with a very small bias.

3.2 Bernoulli distribution

In this subsection, we consider binary distributed data as an example of discrete longitudinal data. Like the normal distribution case, we also look at two cases, i.e., time-independent covariates and time-dependent covariates.

3.2.1 Time-independent covariates

The setting mechanism of the model for Bernoulli data with time-independent covariates is similar to the one for the normal distribution with time-independent covariates. The only difference here lies in that the responses y_{ij} are generated

Table 4 Random effects selection

Method	MRME	No. of zeros		Proportion of		
		C	IC	Under-fit	Correct-fit	Over-fit
SCAD	0.027195	6.87	0.00	0.00	0.88	0.12
LASSO	0.016400	6.87	0.00	0.00	0.88	0.12

Table 5 Random effects selection

Method	MRME	No. of zeros		Proportion of		
		C	IC	Under-fit	Correct-fit	Over-fit
SCAD	4.0e-5	3.91	0.00	0.00	0.91	0.09
LASSO	4.2e-5	3.87	0.00	0.00	0.87	0.13

Table 6 Random effects selection

Method	MRME	No. of zeros		Proportion of		
		C	IC	Under-fit	Correct-fit	Over-fit
SCAD	0.016264	6.96	0.00	0.00	0.97	0.03
LASSO	0.073532	6.93	0.00	0.00	0.94	0.06

according to Bernoulli distributions $\mathrm{Bin}(p_{ij})$, where the success probability p_{ij} is modeled via $\mathrm{logit}(p_{ij}) = \mathrm{logit}(\mu_{ij}) = x_{ij}^T\beta + z_{ij}^T b_i$. Still, x_{ij} are simulated from multivariate-normal distributions with $\rho = 0.5$ for moderate correlation and $\beta = (10, 9, 8, 7, 6, 5, 4, 3, 2, 1)$, The covariance matrix of the random effects $b_i = (b_{i1}, b_{i2}, \ldots, b_{i,10})$ is set to be $D = \mathrm{diag}(3, 2, 1, 0, \ldots, 0)$.

Table 4 is presented to assess the performance of the random effects selection. In Table 4, there are 12 % over-fit using either the SCAD or LASSO penalty method, implying that the proposed method seems to choose some ineffective random effects into the model for longitudinal binary data, but the error is relatively small.

3.2.2 Time-dependent covariates

Using the similar model settings as in the case of normal distribution with time-dependent covariates, we conduct a simulation study with $n = 100$ and $n_i = 10$ for time-dependent covariates case, where the responses y_{ij} are binary distributed with the corresponding success probability $\mathrm{logit}(p_{ij}) = \mathrm{logit}(\mu_{ij}) = x_{ij}^T\beta + z_{ij}^T b_i$.

Table 5 gives a summary of the random effects selection for the model we considered. It is clear that both the LASSO and SCAD methods provide a high rate of correctly selected zero variance components of random effects, though not as high as in the normal distribution case. In contrast to the normal distribution case, the main difference lies in the rate of under- and over-fit. The correct-fit rate is not as high as for the normal distribution case due to the increasing rate of over-fit.

3.3 Binomial distribution

In this subsection, binomial distributed simulation studies are considered as other examples for longitudinal discrete data. The settings of the models are similar to Bernoulli cases, except that this time y_{ij} come from binomial distributions $\mathrm{B}(m, p_{ij})$ where m is the number of trials and p_{ij} is the success probability of the trial. Time-independent covariates and time-dependent covariates cases are considered, respectively.

3.3.1 Time-independent covariates

For the case of time-independent covariates, the response y_{ij} are generated according to binomial distribution $\mathrm{B}(m, p_{ij})$ with corresponding success probability $\mathrm{logit}(p_{ij}) = x_{ij}^T\beta + z_{ij}^T b_i$ and the trial number $m = 4$, where the covariates x_{ij} are generated from multi-normal distribution with $\rho = 0.5$. The true values of the mean parameters β and the covariance matrix of random effects are the same as those in the two previous cases with time-independent covariates.

Table 6 gives the performance of the proposed method for the random effects selection. We note that there is a significant improvement over the Bernoulli case in terms of the correct-fit rate. The SCAD method outperforms the LASSO in terms random effects selection for the binomial case as the former has a higher rate of correct fit and a lower rate of over-fit as well.

3.3.2 Time-dependent covariates

The parameter settings and data generating mechanism are the same as for the case of Bernoulli data with time-

Table 7 Random effects selection

Method	MRME	No. of zeros		Proportion of		
		C	IC	Under-fit	Correct-fit	Over-fit
SCAD	0.017954	3.96	0.00	0.00	0.96	0.04
LASSO	0.017325	3.97	0.00	0.00	0.97	0.03

Table 8 Random effects selection

Method	MRME	No. of zeros		Proportion of		
		C	IC	Under-fit	Correct-fit	Over-fit
SCAD	0.000647	6.98	0.00	0.00	0.98	0.02
LASSO	0.001998	6.95	0.00	0.00	0.96	0.04

Table 9 Random effects selection

Method	MRME	No. of zeros		Proportion of		
		C	IC	Under-fit	Correct-fit	Over-fit
SCAD	2.43e-5	3.99	0.00	0.00	0.99	0.01
LASSO	2.47e-5	3.98	0.00	0.00	0.98	0.02

dependent covariates. The only difference is that the responses y_{ij} are generated from a binomial distribution with the trial number $m = 4$. Table 7 gives the performance of random effects selection. Like in the other cases, the proposed method for random effects selection works very well in the case of time-dependent covariates for binomial data. In terms of comparison between two discrete data cases, binomial data cases outperform Bernoulli cases, which is natural as binomial data provide more information.

3.4 Poisson distribution

In this subsection, we conduct simulation studies for Poisson distributed data, i.e., the responses y_{ij} are Poisson distributed with mean μ_{ij}. Similar to the previous subsections, we also consider two cases of time-independent and time-dependent covariates.

3.4.1 Time-independent covariates

The general parameter settings and data generating mechanism are similar to the cases of normal, Bernoulli and binomial data with time-independent covariates. But this time the responses y_{ij} are generated according to Poisson distributions $\text{Pois}(\mu_{ij})$, where $\log(\mu_{ij}) = x_{ij}^T\beta + z_{ij}^T b_i$ and the covariates x_{ij} and z_{ij} are generated from a multivariate normal distribution with $\rho = 0.5$. Assume $\beta = (1.0, 0.9, \ldots, 0.1)$. Again, the covariance matrix of the random effects $b_i = (b_{i1}, b_{i2}, \ldots, b_{i,10})$ are set as $D = \text{diag}(3, 2, 1, 0, \ldots, 0)$. The simulation results are summarized in Table 8.

Table 8 indicates that for Poisson distribution data both SCAD and LASSO methods have done a good job in selecting effective random effects, as both have very high rates of correct-fit and correct identification of ineffective random effects. In comparison, the SCAD method performs slightly better than the LASSO approach in selecting effective random effects for Poisson data. In addition, it seems that for the Poisson distribution case the proposed random effects selection method improves in correct-fit rate when compared to the performance for Bernoulli/binomial distribution cases.

3.4.2 Time-dependent covariates

Similar to the previous cases with time-dependent covariates, the responses y_{ij} are generated from Poisson distributions with mean μ_{ij}, formed by $\log(\mu_{ij}) = x_{ij}^T\beta + z_{ij}^T b_i$, where x_{ij} and z_{ij} are polynomials in time. The covariance matrix of random effects b_i is taken as $D = \text{diag}(2, 1, 0, 0, 0, 0)$ and the fixed effects are set to be $\beta = (0.6, -0.5, 0.4, -0.3, 0.2, -0.1)$. The simulation results are summarized in Table 9. From Table 9, it is clear that the proposed random effects selection method performs very well for Poisson distribution data with time-dependent covariates.

4 Real data analysis for CD4+ cell data

In this section, we illustrate the use of our proposed method to the analysis of CD4+ data for a HIV study (Kaslow et al. 1987). In total, there were 369 patients enrolled in the study and for each patient repeated measurements were taken. The study period lasted about eight and half years and 2376 measurements are taken in total. The measurement time and

Table 10 Mean parameters estimation with LASSO and SCAD

	β_1	β_2	β_3	β_4	β_5	β_6	β_7
YP06	875.22	−207.27	−22.48	32.21	−1.31	−1.92	0.25
SCAD	874.74	−203.16	−19.33	31.65	−1.37	−1.84	0.24
LASSO	874.75	−203.17	−19.35	31.65	−1.37	−1.84	0.24

measurement number for each patient varied from patient to patient, so that the data set is highly unbalanced. This CD4+ data has already shown a very strong individual variability (Kaslow et al. 1987), leading to the use of GLMM to accommodate individual heterogeneities.

The CD4+ data has been analyzed by many authors including Groll and Tutz (2012) and Ye and Pan (2006) among others. In Ye and Pan (2006), the mean and variance for the CD4+ data were modeled in terms of polynomials in time, but random effects were not considered. Below we use the same polynomial in time as Ye and Pan (2006) to model the mean structure but add potential random effects, here up to random cubic term, into the model. In other words,

$$y_{ij} = \beta_1 + \beta_2 t_{ij} + \beta_3 t_{ij}^2 + \beta_4 t_{ij}^3 + \beta_5 t_{ij}^4 + \beta_6 t_{ij}^5 + \beta_7 t_{ij}^6 + b_{i1} + b_{i2} t_{ij} + b_{i3} t_{ij}^2 + b_{i4} t_{ij}^3 + \epsilon_{ij},$$

where y_{ij} is the CD4+ cell number of the ith patient measured at jth time point, $b_i = (b_{i1}, b_{i2}, b_{i3}, b_{i4})^T$ are the potential random effects following a normal distribution with mean 0 and covariance matrix D, and ϵ_{ij} are random errors following a normal distribution.

Table 10 presents the estimates of β using SCAD and LASSO methods and Ye and Pan (2006)'s estimation results. In Table 10, 'YP06' represents the estimates of the fixed effects β by Ye and Pan (2006) and the other results are those of our proposed method using LASSO and SCAD penalties. From Table 10, we can see that the mean parameter estimates using the proposed method are quite close to those of Ye and Pan (2006).

For the selection of random effects, we make a comparison between the proposed method and a full screening search method based on AIC or BIC selection criterion. At the initial stage, a polynomial in time with likely random effects up to random cubic term is taken as the starting model, as shown in the model above. The random effects b_{i1}, b_{i2}, b_{i3}, b_{i4} may be all needed or may not be needed, so that there are $2^4 - 1 = 15$ possible sub-models when incorporating random effects. Based on the criterion of either AIC or BIC, the best three sub-models selected from all the sub-models, which we rank as 1st, 2nd and 3rd models, are presented in Table 11. Although the AIC or BIC-based selection method is computationally intensive, they may select a reasonably good model from all the sub-models. Thus the AIC or BIC-based selected model can be used as a benchmark for comparison purpose. Table 11 reports the selected random effects using the AIC or BIC model selection method and our proposed penalty-based method, respectively. Table 11 shows that our proposed SCAD- or LASSO-penalty selection method selects b_{i1}, b_{i4} as the effective random effects, which is consistent with the ones selected by using the AIC or BIC-based screening search method.

Table 11 Random effects selection

Methods	Rank	Selected random effects
AIC	1st	b_{i1}, b_{i4}
	2nd	b_{i1}, b_{i2}, b_{i3}
	3rd	b_{i1}, b_{i2}, b_{i4}
BIC	1st	b_{i1}, b_{i4}
	2nd	b_{i1}, b_{i2}, b_{i3}
	3rd	b_{i1}, b_{i2}, b_{i4}
SCAD		b_{i1}, b_{i4}
LASSO		b_{i1}, t_{i4}

Note that the AIC or BIC-based screening search method requires to estimate the parameters in each sub-model and to make comparison for all possible sub-models. Inevitably, it is very computationally intensive, in particular when the dimension of random effects q is large, as there are $2^q - 1$ possible random effects sub-models to consider. In contrast, our proposed approach only requires to work on one model, which is able to simultaneously select effective random effects and estimate parameters in the model.

Finally, we report that in our analysis the estimates of the variances of the effective random effects b_{i1} and b_{i4} are, respectively, 2.61 and 0.04 by the LASSO method, and 3.51 and 0.04 by the SCAD method. Similar to Groll and Tutz (2013), in Fig. 3 we provide a graphical illustration for variance build-ups of random effects for CD4+ cell count data using SCAD penalty. From Fig. 3, it is clear that the variances estimates of $b_{i1}, \ldots, b_{i4}$ are all not zero at $\psi = 0.0$, but at the optimal tuning parameter $\psi = 1.0$ the variance estimates of b_{i2} and b_{i3} become zero. Note for b_{i4} it is 0.04 at $\psi = 1.0$, which is close but not equal to zero. For illustration, in Fig. 4 we display the estimated population mean (solid curve), and the estimated individual trajectory for Subject 329 (dashed curve) together with his CD4+ cell count data (circle points), based on the model including ran-

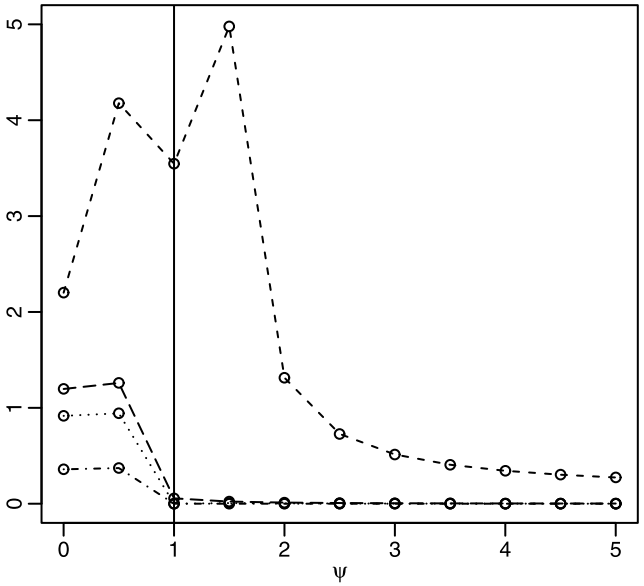


Fig. 3 Random effects variances buildups for CD4+ cell count data; the optimal value of the tuning parameter ψ is shown by the *vertical line*

dom effects b_{i1} and b_{i4} using LASSO and SCAD penalties. It shows that the proposed random effects models using both LASSO and SCAD penalties fit the data reasonably well.

5 Discussions

The PQL estimation method has been widely used in GLMMs due to its easy computation, although some bias may likely occur in the estimation of the variance components, in particular when the data are binary or the sample size is small. For binary and binomial correlated data, Breslow and Lin (1995) and Lin and Breslow (1996) showed that the PQL estimates of the parameters in GLMMs are acceptable if the data are not too sparse. For Poisson correlated data, the PQL estimation has a smaller bias than those in the binomial case (Lin 2007). With a normal distribution assumption, GLMMs reduce to LMMs and in this case the PQL approach works reasonably well, so that the proposed SCAD or LASSO penalty method performs very well in selecting effective random effects and estimating variance components, as we saw in our simulation studies. For GLMMs, the method proposed in this paper also works well in selecting effective random effects, although the estimates of the variance components using the ML version (2.5) may have some bias. Note that such bias can be reduced by using the REML version (2.6) when selecting effective random effects and estimating the fixed effects and variance components, simultaneously.

As we mentioned in Sect. 1, there is already some literature research on variable selection for fixed effects in

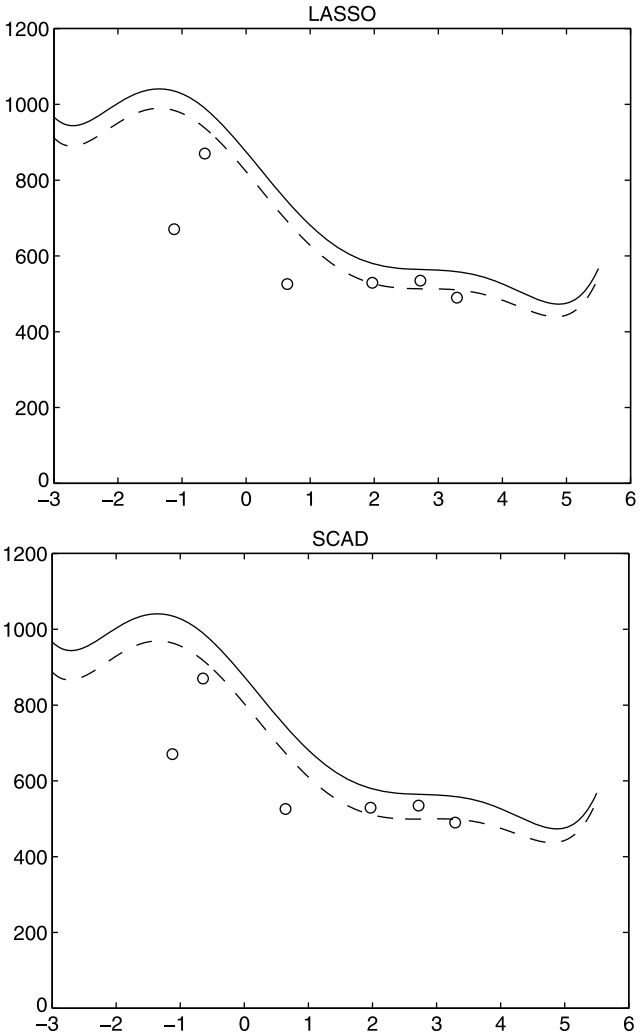


Fig. 4 Population and individual mean *curves* with original data of a certain subject

GLMMs. In this paper, we focus on selecting effective random effects by imposing either SCAD or LASSO penalty functions of the variance components of random effects to the PQL function. An extension of the proposed approach to selection of both fixed effects and random effects, simultaneously, is rather straightforward. For example, this can be done by adding a second penalty function of the fixed effects β to the penalized log-likelihood function (2.5) or (2.6), making the profile log-likelihood function $\ell_2\,(\hat{\beta}, \theta)$ or $\ell_{2R}\,(\hat{\beta}, \theta)$ be penalized doubly. We will report such studies in our follow-up paper.

Acknowledgements Pan's research was supported by a grant from the Royal Society of the UK, and Huang's research was funded by a scholarship from the University of Manchester. We would like to thank two anonymous referees and the Editor/AE for their constructive comments and helpful suggestions.

Appendix: Derivation of the first- and second-order derivatives

Part 1. First we show how to derive $\Sigma_\psi(\theta_1^{(k-1)})$ and $U_\psi(\theta_1^{(k-1)})$. The main idea is to localize $p_\psi(|\theta_1|)$ approximately by a quadratic function. This is a direct result of Fan and Li (2001). Suppose that we already have $\theta_1^{(k-1)} = (\lambda_1^{(k-1)}, \ldots, \lambda_q^{(k-1)})$ that is close to the minimizer of (2.5). Then $p_\psi(|\lambda_l|), l = 1, \ldots, q$ can be locally approximated by a quadratic function as

$$\big[p_\psi\big(|\lambda_l|\big)\big]' = p'_\psi\big(|\lambda_l|\big)\,\mathrm{sgn}(\lambda_l) \approx \frac{p'_\psi(|\lambda_l^{(k-1)}|)}{|\lambda_l^{(k-1)}|}\lambda_l$$

for $\lambda_l \approx \lambda_l^{(k-1)}$, where $\mathrm{sgn}(\lambda_l)$ is the sign of λ_l. In other words,

$$p_\psi\big(|\lambda_l|\big) \approx p_\psi\big(|\lambda_l|^{(k-1)}\big) + \frac{1}{2}\frac{p'_\psi(|\lambda_l^{(k-1)}|)}{|\lambda_l^{(k-1)}|}\big(\lambda_l^2 - \big(\lambda_l^{(k-1)}\big)^2\big).$$

Thus, with the local approximated form of $p_\psi(|\theta_1|) = (p_\psi(|\lambda_1|), \ldots, p_\psi(|\lambda_q|))$, we can derive its second and first derivatives, i.e., $\Sigma_\psi(\theta_1^{(k-1)})$ and $U_\psi(\theta_1^{(k-1)})$ at given value $\theta_1^{(k-1)}$.

Part 2. Without lose of generality, we suppress $(k-1)$ in $\theta_i^{(k-1)}$. We will show how to get the elements in $\nabla l_2(\theta_i)$ and $\nabla^2 l_2(\theta_i)$, and similarly in $\nabla l_{2R}(\theta_i)$ and $\nabla^2 l_{2R}(\theta_i)$, $i = 1, 2$, given $\theta_1 = (\lambda_1, \ldots, \lambda_q)$ and $\theta_2 = (\gamma_{21}; \gamma_{31}, \gamma_{32}; \ldots; \gamma_{q1}, \ldots, \gamma_{q(q-1)})$.

With $l_2(\hat\beta, \theta) = -\frac{1}{2}\sum_{i=1}^n log|V_i| - \frac{1}{2}\sum_{i=1}^n (Y_i - X_i\hat\beta)^T V_i^{-1}(Y_i - X_i\hat\beta)$ and let $e_i = Y_i - X_i\hat\beta$, we have

$$l_2(\hat\beta, \theta) = -\frac{1}{2}\sum_{i=1}^n \log|V_i| - \frac{1}{2}\sum_{i=1}^n e_i^T V_i^{-1} e_i. \tag{A.1}$$

Similarly, we have

$$l_{2R}(\hat\beta, \theta) = -\frac{1}{2}\sum_{i=1}^n \log|V_i| - \frac{1}{2}\sum_{i=1}^n e_i^T V_i^{-1} e_i - \frac{1}{2}\sum_{i=1}^n \log\big|X_i^T V_i^{-1} X_i\big|. \tag{A.2}$$

At first we deal with the elements in $\nabla l_2(\theta_1)$ and $\nabla^2 l_2(\theta_1)$, i.e., $\frac{\partial l_2(\hat\beta,\theta)}{\partial\lambda_l}$ and $\frac{\partial l_2(\hat\beta,\theta)}{\partial\lambda_l\partial\lambda_m}$, where $l, m = 1, \ldots, q$. From (A.1),

$$\frac{\partial l_2(\hat\beta,\theta)}{\partial\lambda_l} = \sum_{k=1}^n\left\{-\frac{1}{2}\mathrm{tr}\left(V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}\right) + \frac{1}{2}e_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1}e_k\right\},$$

$$\frac{\partial l_2(\hat\beta,\theta)}{\partial\lambda_l\partial\lambda_m} = \sum_{k=1}^n\left\{\frac{1}{2}\mathrm{tr}\left(V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1}\frac{\partial V_k}{\partial\lambda_m}\right) - \frac{1}{2}\mathrm{tr}\left(V_k^{-1}\frac{\partial^2 V_k}{\partial\lambda_l\partial\lambda_m}\right) - e_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1}\frac{\partial V_k}{\partial\lambda_m}V_k^{-1}e_k + \frac{1}{2}e_k^T V_k^{-1}\frac{\partial^2 V_k}{\partial\lambda_l\partial\lambda_m}V_k^{-1}e_k\right\}.$$

Similarly we can calculate the elements in $\nabla l_{2R}(\theta_1)$ and $\nabla^2 l_{2R}(\theta_1)$, i.e., $\frac{\partial l_{2R}(\hat\beta,\theta)}{\partial\lambda_l}$ and $\frac{\partial l_{2R}(\hat\beta,\theta)}{\partial\lambda_l\partial\lambda_m}$, where $l, m = 1, \ldots, q$. From (A.2),

$$\frac{\partial l_{2R}(\hat\beta,\theta)}{\partial\lambda_l} = \sum_{k=1}^n\left\{-\frac{1}{2}\mathrm{tr}\left(V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}\right) + \frac{1}{2}e_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1}e_k + \frac{1}{2}\mathrm{tr}\left(\big(X_k^T V_k^{-1}X_k\big)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1}X_k\right)\right\}$$

$$\begin{aligned}\frac{\partial l_{2R}(\hat\beta,\theta)}{\partial\lambda_l\partial\lambda_m} = \sum_{k=1}^n\Bigg\{&\frac{1}{2}\mathrm{tr}\left(V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1}\frac{\partial V_k}{\partial\lambda_m}\right) - \frac{1}{2}\mathrm{tr}\left(V_k^{-1}\frac{\partial^2 V_k}{\partial\lambda_l\partial\lambda_m}\right) - e_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1}\frac{\partial V_k}{\partial\lambda_m}V_k^{-1}e_k + \frac{1}{2}e_k^T V_k^{-1}\frac{\partial^2 V_k}{\partial\lambda_l\partial\lambda_m}V_k^{-1}e_k\\ &- \frac{1}{2}\mathrm{tr}\left(\big(X_k^T V_k^{-1}X_k\big)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1}X_k \times \big(X_k^T V_k^{-1}X_k\big)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_m}V_k^{-1}X_k\right)\\ &- \frac{1}{2}\mathrm{tr}\left(\big(X_k^T V_k^{-1}X_k\big)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_l}V_k^{-1} \times \frac{\partial V_k}{\partial\lambda_m}V_k^{-1}X_k\right)\\ &- \frac{1}{2}\mathrm{tr}\left(\big(X_k^T V_k^{-1}X_k\big)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial\lambda_m}V_k^{-1} \times \frac{\partial V_k}{\partial\lambda_l}V_k^{-1}X_k\right)\\ &+ \frac{1}{2}\mathrm{tr}\left(\big(X_k^T V_k^{-1}X_k\big)^{-1}X_k^T V_k^{-1} \times \frac{\partial^2 V_k}{\partial\lambda_l\partial\lambda_m}V_k^{-1}X_k\right)\Bigg\}\end{aligned}$$

For the derivatives with respect to θ_2, we have similar expressions. Denote the elements in $\nabla l_2(\theta_2)$ by $\frac{\partial l_2(\hat\beta,\theta)}{\partial\gamma_{s_1 r_1}}$ where $s_1 = 2, \ldots, q$; $r_1 = 1, \ldots, l_1 - 1$. And denote the elements in $\nabla^2 l_2(\theta_2)$ by $\frac{\partial l_2(\hat\beta,\theta)}{\partial\gamma_{s_1 r_1}\partial\gamma_{s_2 r_2}}$, where $s_2 = 2, \ldots, q$; $r_2 = 1, \ldots, l_2 - 1$. From (A.1),

$$\frac{\partial l_2(\hat{\beta},\theta)}{\partial \gamma_{s_1r_1}} = \sum_{k=1}^{n}\Bigg\{-\frac{1}{2}\mathrm{tr}\Bigg(V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}\Bigg) + \frac{1}{2}e_k^T V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}e_k\Bigg\}$$

$$\begin{aligned}\frac{\partial l_2(\hat{\beta},\theta)}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}} = \sum_{k=1}^{n}\Bigg\{&\frac{1}{2}\mathrm{tr}\Bigg(V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_2r_2}}\Bigg) - \frac{1}{2}\mathrm{tr}\Bigg(V_k^{-1}\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}}\Bigg)\\ &- e_k^T V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_2r_2}}V_k^{-1}e_k + \frac{1}{2}e_k^T V_k^{-1}\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}}V_k^{-1}e_k\Bigg\}\end{aligned}$$

Similarly form (A.2), we have

$$\begin{aligned}\frac{\partial l_{2R}(\hat{\beta},\theta)}{\partial \gamma_{s_1r_1}} = \sum_{k=1}^{n}\Bigg\{&-\frac{1}{2}\mathrm{tr}\Bigg(V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}\Bigg) + \frac{1}{2}e_k^T V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}e_k\\ &+ \frac{1}{2}\mathrm{tr}\Bigg(\left(X_k^T V_k^{-1}X_k\right)^{-1}X_k^T V_k^{-1}\times\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}X_k\Bigg)\Bigg\}\end{aligned}$$

$$\begin{aligned}\frac{\partial l_{2R}(\hat{\beta},\theta)}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}} = \sum_{k=1}^{n}\Bigg\{&\frac{1}{2}\mathrm{tr}\Bigg(V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_2r_2}}\Bigg) - \frac{1}{2}\mathrm{tr}\Bigg(V_k^{-1}\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}}\Bigg)\\ &- e_k^T V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_2r_2}}V_k^{-1}e_k + \frac{1}{2}e_k^T V_k^{-1}\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}}V_k^{-1}e_k\\ &- \frac{1}{2}\mathrm{tr}\Bigg(\left(X_k^T V_k^{-1}X_k\right)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}\times X_k\left(X_k^T V_k^{-1}X_k\right)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_2r_2}}V_k^{-1}X_k\Bigg)\\ &- \frac{1}{2}\mathrm{tr}\Bigg(\left(X_k^T V_k^{-1}X_k\right)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}\times\frac{\partial V_k}{\partial \gamma_{s_2r_2}}V_k^{-1}X_k\Bigg)\\ &- \frac{1}{2}\mathrm{tr}\Bigg(\left(X_k^T V_k^{-1}X_k\right)^{-1}X_k^T V_k^{-1}\frac{\partial V_k}{\partial \gamma_{s_2r_2}}V_k^{-1}\times\frac{\partial V_k}{\partial \gamma_{s_1r_1}}V_k^{-1}X_k\Bigg)\\ &+ \frac{1}{2}\mathrm{tr}\Bigg(\left(X_k^T V_k^{-1}X_k\right)^{-1}X_k^T V_k^{-1}\times\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}}V_k^{-1}X_k\Bigg)\Bigg\}\end{aligned}$$

Finally, we give the expression forms of $\frac{\partial V_k}{\partial \lambda_l}$ ($\frac{\partial V_k}{\partial \lambda_m}$), $\frac{\partial^2 V_k}{\partial \lambda_l \partial \lambda_m}$ and $\frac{\partial V_k}{\partial \gamma_{s_1r_1}}$ ($\frac{\partial V_k}{\partial \gamma_{s_2r_2}}$), $\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}}$ in the above equations. First we have

$$\begin{aligned}\frac{\partial V_k}{\partial \lambda_l} &= \frac{\partial(Z_k \Lambda\Gamma\Gamma^T\Lambda Z_k^T + W_k^{-1})}{\partial \lambda_l}\\ &= Z_k\frac{\partial \Lambda}{\partial \lambda_l}\Gamma\Gamma^T\Lambda Z_k^T + Z_k\Lambda\Gamma\Gamma^T\frac{\partial \Lambda}{\partial \lambda_l}Z_k^T,\end{aligned}$$

where $\frac{\partial \Lambda}{\partial \lambda_l}$ is a $q\times q$ matrix with 1 at the lth diagonal entry and 0 at all the other entries. Also we have

$$\frac{\partial^2 V_k}{\partial \lambda_l \partial \lambda_m} = Z_k\frac{\partial \Lambda}{\partial \lambda_l}\Gamma\Gamma^T\frac{\partial \Lambda}{\partial \lambda_m}Z_k^T + Z_k\frac{\partial \Lambda}{\partial \lambda_m}\Gamma\Gamma^T\frac{\partial \Lambda}{\partial \lambda_l}Z_k^T,$$

where $\frac{\partial \Lambda}{\partial \lambda_m}$ is a $q\times q$ matrix with 1 at the mth diagonal entry and 0 at all the other entries.

If $l\neq m$, it is easy to see that $\frac{\partial^2 V_k}{\partial \lambda_l \partial \lambda_m}$ becomes to a $q\times q$ matrix with 0 at all entries; if $l=m$, $\frac{\partial^2 V_k}{\partial \lambda_l \partial \lambda_m} = 2 * Z_k U_l Z_k^T$, where U_l is a $q\times q$ matrix with $1+\Sigma_{i=1}^{l-1}\gamma_{li}^2$ at the lth diagonal entry and 0 at all the other entries.

The expression form of $\frac{\partial V_k}{\partial \gamma_{s_1r_1}}$ is

$$\begin{aligned}\frac{\partial V_k}{\partial \gamma_{s_1r_1}} &= \frac{\partial(Z_k \Lambda\Gamma\Gamma^T\Lambda Z_k^T + W_k^{-1})}{\partial \gamma_{s_1r_1}}\\ &= Z_k\Lambda\frac{\partial \Gamma\Gamma^T}{\partial \gamma_{s_1r_1}}\Lambda Z_k^T\\ &= Z_k\Lambda\frac{\partial \Gamma}{\partial \gamma_{s_1r_1}}\Gamma^T\Lambda Z_k^T + Z_k\Lambda\Gamma\left(\frac{\partial \Gamma}{\partial \gamma_{s_1r_1}}\right)^T\Lambda Z_k^T,\end{aligned}$$

where $\frac{\partial \Gamma}{\partial \gamma_{s_1r_1}}$ is a $q\times q$ matrix with 1 at the s_1th row and r_1th column entry and 0 at all the other entries. Furthermore, we have

$$\begin{aligned}\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}} &= Z_k\Lambda\frac{\partial \Gamma}{\partial \gamma_{s_1r_1}}\left(\frac{\partial \Gamma}{\partial \gamma_{s_2r_2}}\right)^T\Lambda Z_k^T\\ &\quad + Z_k\Lambda\frac{\partial \Gamma}{\partial \gamma_{s_2r_2}}\left(\frac{\partial \Gamma}{\partial \gamma_{s_1r_1}}\right)^T\Lambda Z_k^T\end{aligned}$$

If $r_1\neq r_2$, $\frac{\partial \Gamma}{\partial \gamma_{s_1r_1}}(\frac{\partial \Gamma}{\partial \gamma_{s_2r_2}})^T$ turns to be a zero matrix, hence $\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}}$ also becomes a zero matrix; if $r_1=r_2$, $\frac{\partial \Gamma}{\partial \gamma_{s_1r_1}}(\frac{\partial \Gamma}{\partial \gamma_{s_2r_2}})^T$ turns into a matrix with 1 at s_1th row and s_2th column entry and 0 at all the other entries. At this time, if $s_1\neq s_2$, $\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}}$ become a zero matrix since Λ is a diagonal matrix; if $s_1=s_2$, $\frac{\partial^2 V_k}{\partial \gamma_{s_1r_1}\partial \gamma_{s_2r_2}} = 2 * Z_k S_{s_1} Z_k^T$, where S_{s_1} is matrix with $\lambda_{s_1}^2$ at the s_1th diagonal entry and 0 at all the other entries.

References

Ahn, M., Zhang, H.H., Lu, W.: Moment-based method for random effects selection in linear mixed models. Stat. Sin. **22**, 1539–1562 (2012)

Bondell, H.D., Krishna, A., Ghosh, S.K.: Joint variable selection for fixed and random effects in linear mixed models. Biometrics **66**, 1069–1077 (2010)

Breiman, L.: Heuristics of instability and stabilization in model selection. Ann. Stat. **24**, 2350–2383 (1996)

Breslow, N.E., Clayton, D.G.: Approximate inference in generalized linear mixed models. J. Am. Stat. Assoc. **88**, 9–25 (1993)

Breslow, N.E., Lin, X.H.: Bias correction in generalised linear mixed models with a single component of dispersion. Biometrika **82**, 81–91 (1995)

Chen, Z., Dunson, D.B.: Random effects selection in linear mixed models. Biometrics **59**, 762–769 (2003)

Fahrmeir, L., Kneib, T., Konrath, S.: Bayesian regularisation in structured additive regression: a unifying perspective on shrinkage, smoothing and predictor selection. Stat. Comput. **20**, 203–219 (2010)

Fan, J.Q., Li, R.Z.: Variable selection via nonconcave penalized likelihood and its oracle properties. J. R. Stat. Assoc. **96**, 1348–1360 (2001)

Fan, Y., Li, R.Z.: Variable selection in linear mixed effects models. Ann. Stat. **40**, 2043–2068 (2012)

Frank, I.E., Friedman, J.H.: A statistical view of some chenometric regression tools (with discussion). Technometrics **35**, 109–148 (1993)

Groll, A., Tutz, G.: Variable selection for generalized linear mixed models by L1-penalized estimation. Stat. Comput. (2013). doi:10.1007/s11222-012-9359-z

Groll, A., Tutz, G.: Regularization for generalized additive mixed models by likelihood-based boosting. Methods Inf. Med. **51**, 168–177 (2012)

Hoerl, A.E., Kennard, R.W.: Ridge regression: biased estimation for nonorthogonal problems. Technometrics **12**, 55–67 (1970a)

Hoerl, A.E., Kennard, R.W.: Ridge regression: application to nonorthogonal problems. Technometrics **12**, 69–82 (1970b)

Ibrahim, J.G., Zhu, H., Garcia, R.I., Guo, R.: Fixed and random effects selection in mixed effects models. Biometrics **67**, 495–503 (2010)

Kaslow, R.A., Ostrow, D.G., Detels, R., et al.: The multicenter AIDS cohort study: rationale, organization and selected characteristics of the participants. Am. J. Epidemiol. **126**, 310–318 (1987)

Kinney, S.K., Dunson, D.B.: Fixed and random effects selection in linear and logistic models. Biometrics **63**, 690–698 (2007)

Lin, B., Pang, Z., Jiang, J.: Fixed and random effects selection by REML and pathwise coordinate optimization. J. Comput. Graph. Stat. (2012). doi:10.1080/10618600.2012.681219

Lin, X.H.: Estimation using penalized quasilikelihood and quasi-pseudo-likelihood in Poisson mixed models. Lifetime Data Anal. **13**, 533–544 (2007)

Lin, X.H., Breslow, N.E.: Bias correction in generalized linear mixed models with multiple components of dispersion. J. Am. Stat. Assoc. **91**, 1007–1016 (1996)

Pan, J., Thompson, R.: Quasi-Monte Carlo estimation in generalized linear mixed models. Comput. Stat. Data Anal. **51**, 5765–5775 (2007)

Patterson, H.D., Thompson, R.: Recovery of inter-block information when block sizes are unequal. Biometrika **58**, 545–554 (1971)

Schall, R.: Estimation in generalized linear models with random effects. Biometrika **78**, 719–727 (1991)

Schelldorfer, J., Bühlmann, P.: GLMMLasso: an algorithm for highdimensional generalized linear mixed models using L1-penalization. Preprint, ETH Zurich (2011). http://stat.ethz.ch/people/schell

Stiratelli, R., Laird, N.M., Ware, J.H.: Random effects models for serial observations with binary response. Biometrics **40**, 961–971 (1984)

Tibshirani, R.J.: Regression shrinkage and selection via the lasso. J. R. Stat. Soc. B **58**, 267–288 (1996)

Ye, H.J., Pan, J.X.: Modelling covariance structures in generalized estimating equations for longitudinal data. Biometrika **93**, 927–941 (2006)

Zeger, S.L., Liang, K., Albert, P.S.: Models for longitudinal data: a generalized estimating equation approach. Biometrics **44**, 1049–1060 (1988)

Zou, H.: The adaptive LASSO and its oracle properties. J. Am. Stat. Assoc. **101**, 1418–1429 (2006)